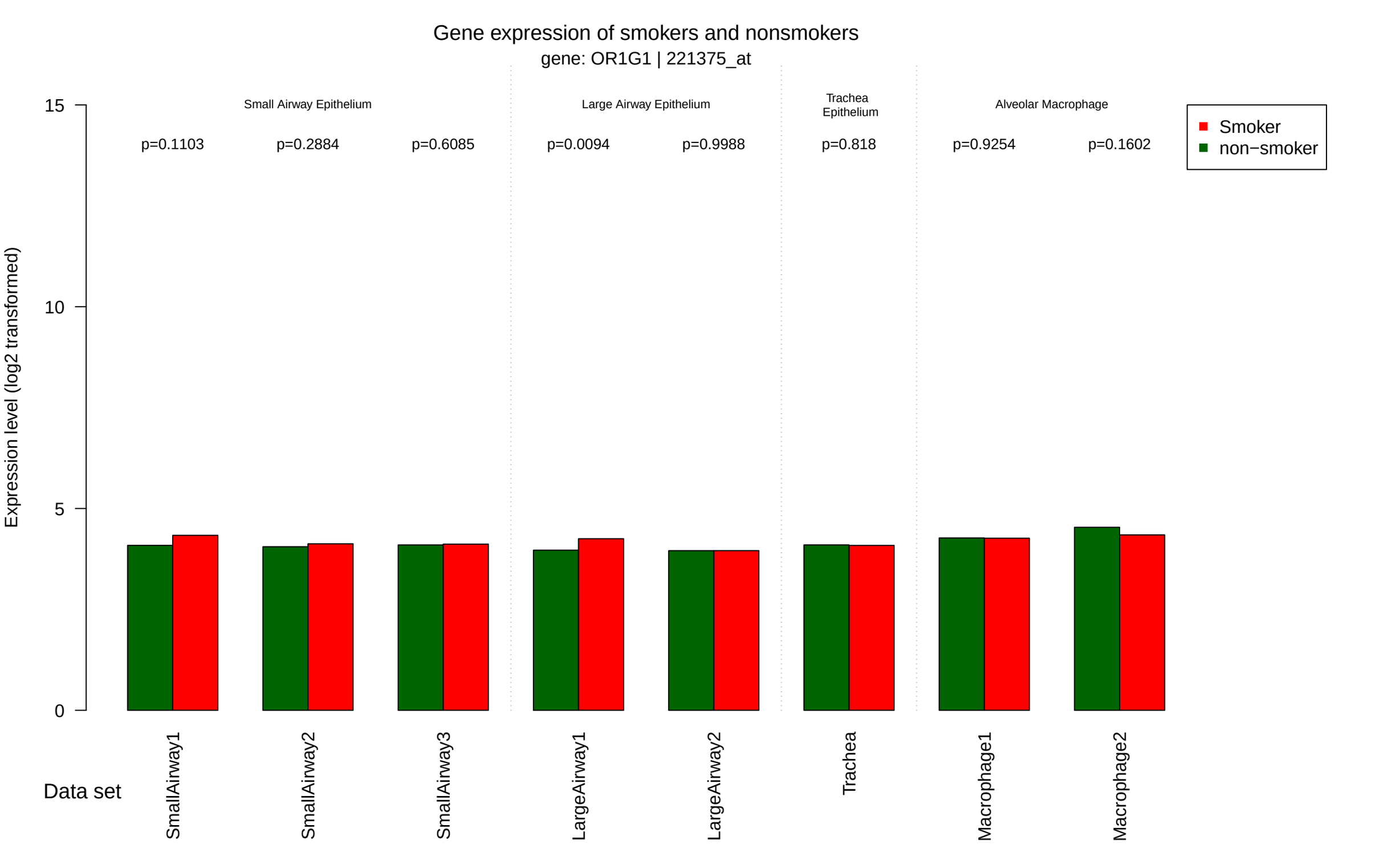
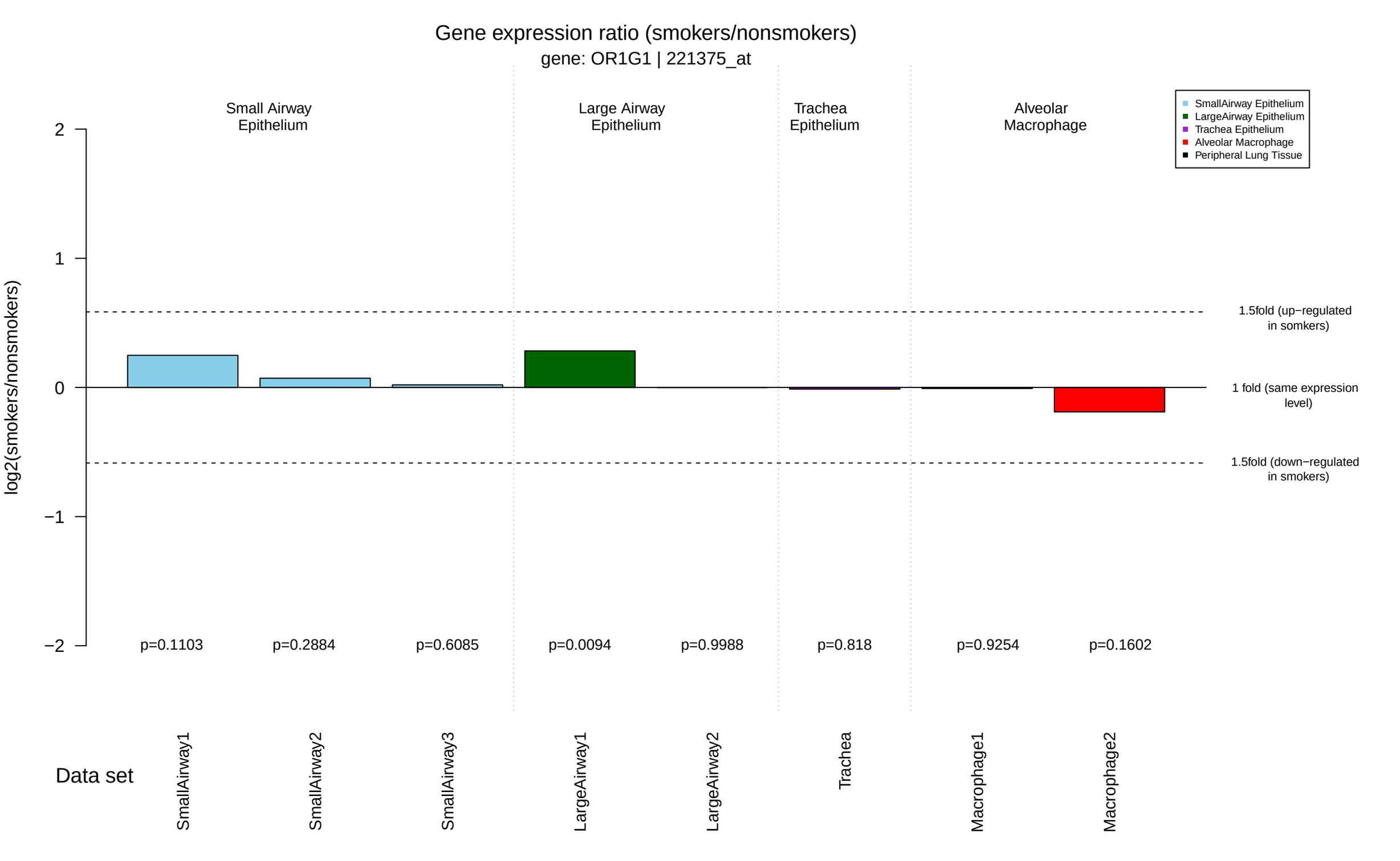
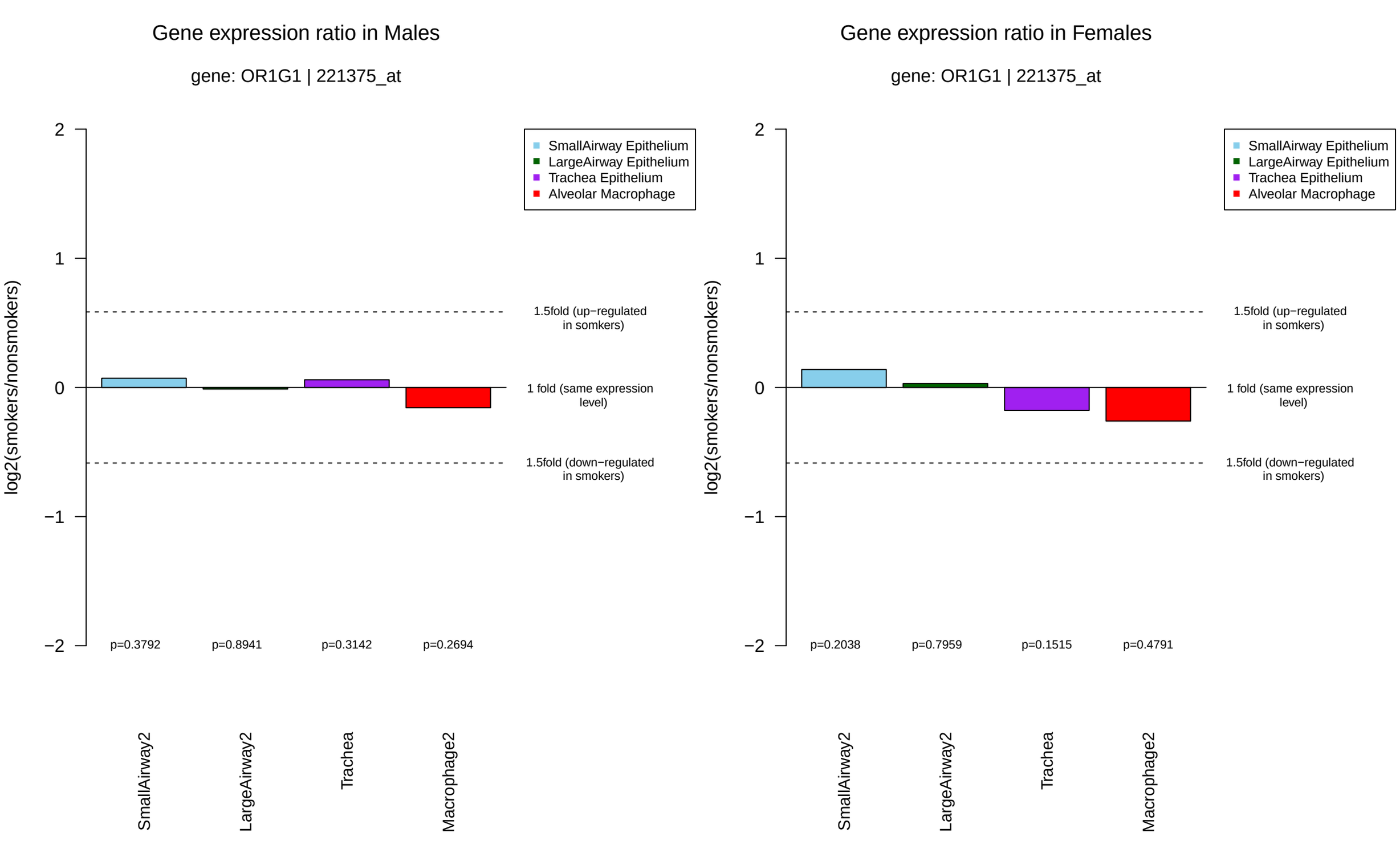
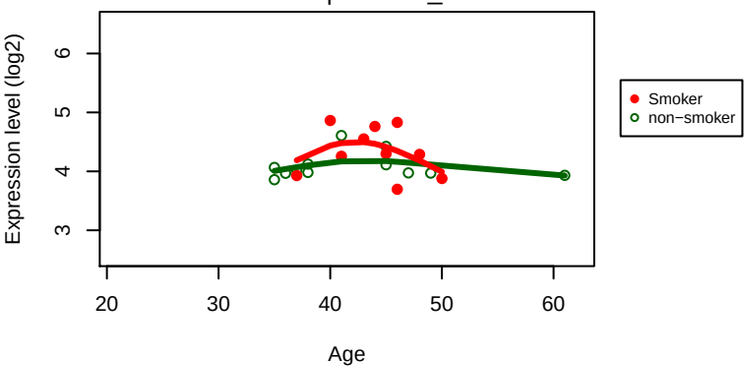
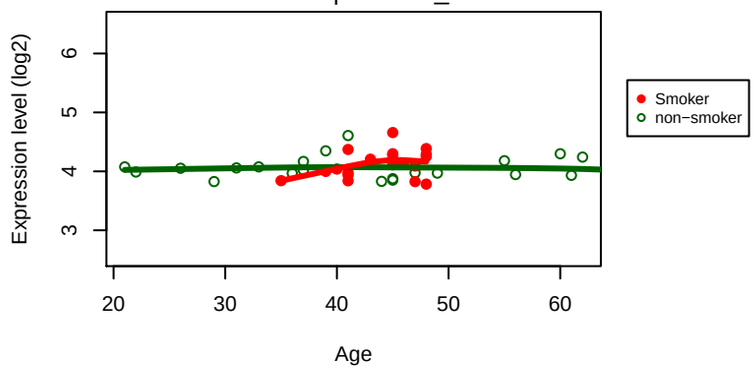
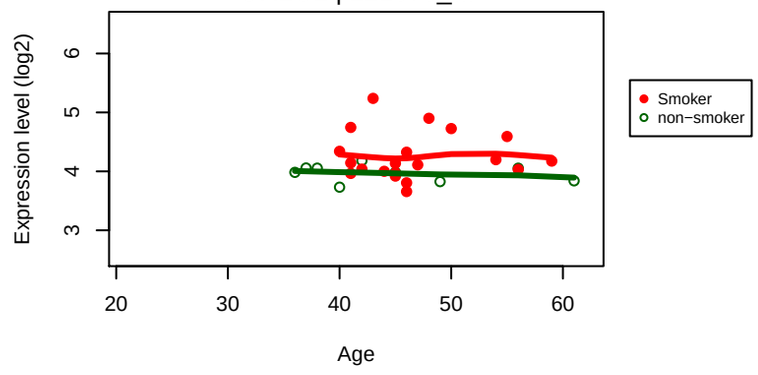

Gene expression of smokers and nonsmokers
gene: OR1G1 | 221375_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1103
p=0.2884
p=0.6085
p=0.0094
p=0.9988
p=0.818
p=0.9254
p=0.1602
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

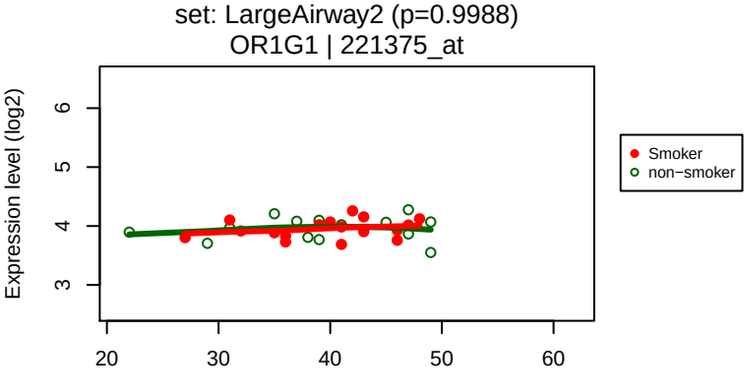

Gene expression ratio (smokers/nonsmokers)
gene: OR1G1 | 221375_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1103
p=0.2884
p=0.6085
p=0.0094
p=0.9988
p=0.818
p=0.9254
p=0.1602
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

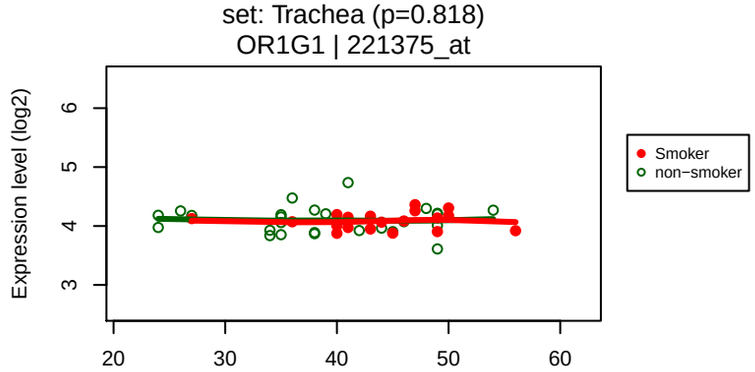

Gene expression ratio in Males
gene: OR1G1 | 221375_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3792
p=0.8941
p=0.3142
p=0.2694
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR1G1 | 221375_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2038
p=0.7959
p=0.1515
p=0.4791
SmallAirway2
LargeAirway2
Trachea
Macrophage2

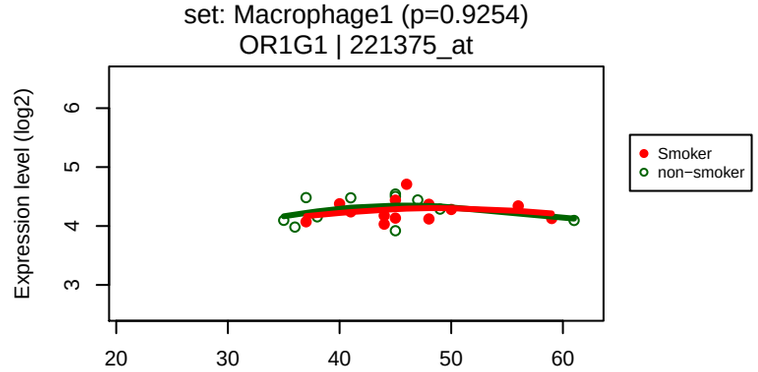

set: SmallAirway1 (p=0.1103)
OR1G1 | 221375_at
Expression level (log2)
Age
Smoker
non−smoker

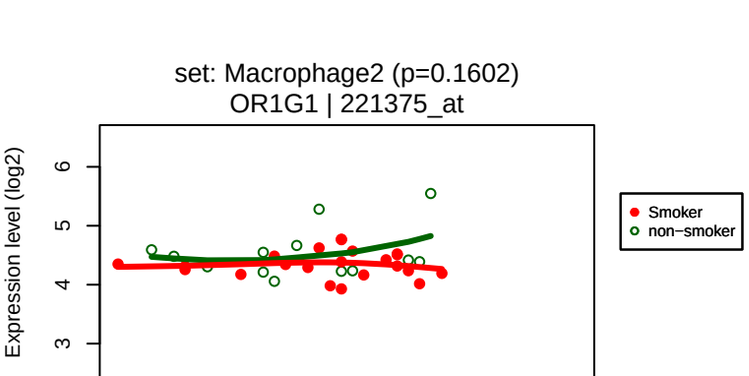

set: SmallAirway2 (p=0.2884)
OR1G1 | 221375_at
Expression level (log2)
Age
Smoker
non−smoker

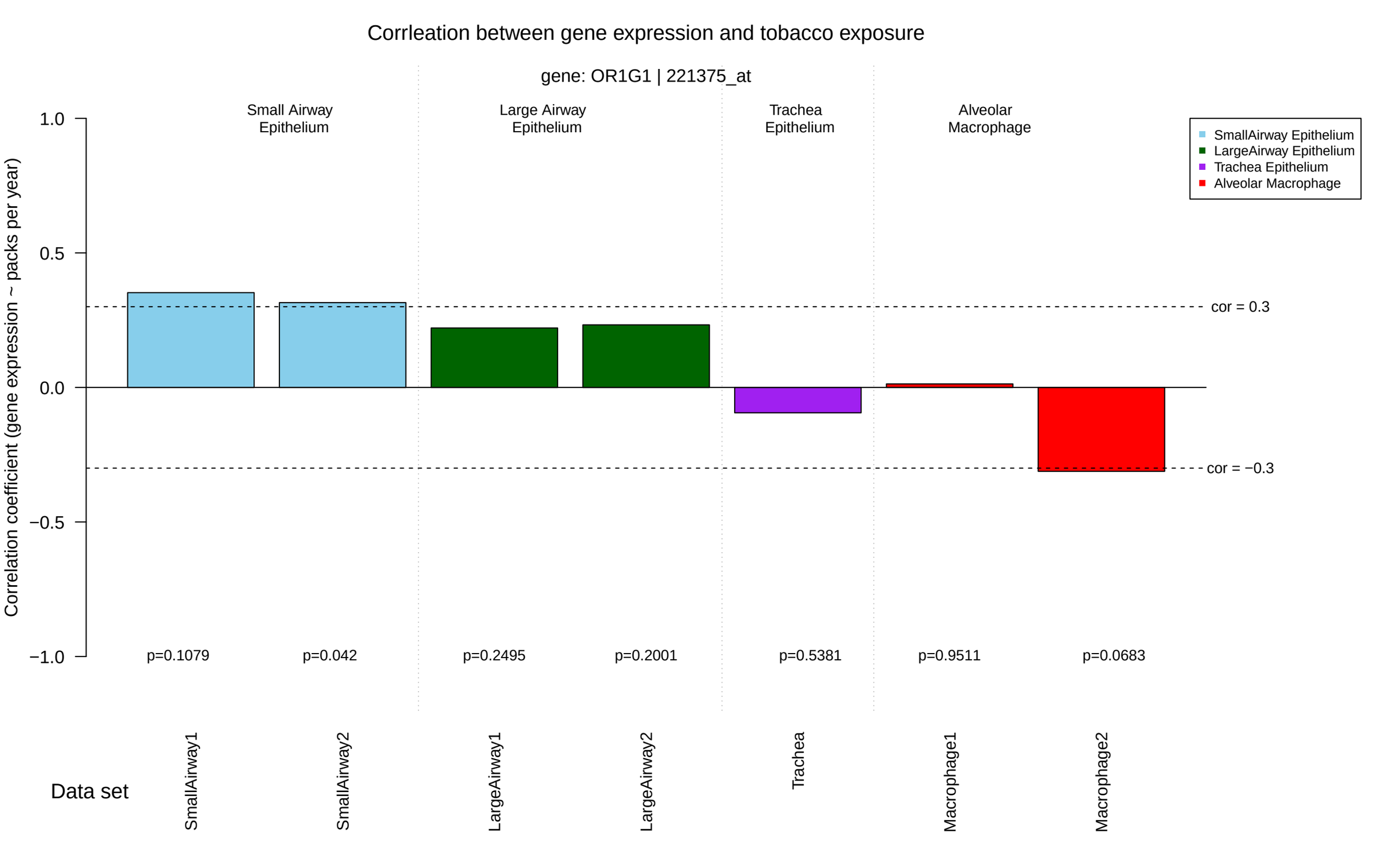

set: LargeAirway1 (p=0.0094)
OR1G1 | 221375_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9988)
OR1G1 | 221375_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.818)
OR1G1 | 221375_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9254)
OR1G1 | 221375_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1602)
OR1G1 | 221375_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR1G1 | 221375_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1079
p=0.042
p=0.2495
p=0.2001
p=0.5381
p=0.9511
p=0.0683
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2